

3. CORE EQUATIONS

The inference process in Collapse-Aware AI is modeled as a stochastic differential equation (SDE):

$$dz_t = b\psi(z_t, M_t) dt + S dW_t$$

The drift term $b\psi$ integrates three informational gradients: memory alignment, anchor stability, and entropy minimization.

The diffusion term $S dW_t$ introduces controlled exploration under uncertainty. Both components are dynamically regulated.

4. COLLAPSE REGIMES

Collapse dynamics exhibit three distinct behavioral regimes:

- Controlled: Strong bias alignment, stable low-entropy collapse.
- Hedge: Rising uncertainty, suppressor activation, softened phrasing.
- Chaos: Drift-dominant collapse leading to hallucinations or loops.

The system detects regime transitions and adjusts parameters to maintain stability.

5. GOVERNOR ARCHITECTURE

The Governor modulates gain coefficients (α , β , γ) at each step based on uncertainty, entropy spikes, and anchor proximity.

It reinforces the Controlled regime, permits recalibration in Hedge, and prevents collapse into Chaos by pulling trajectories toward stable anchors.

6. PHASE-1 VS PHASE-2 IMPLEMENTATION

Phase-1 approximates collapse regulation behaviorally using THB scoring, suppressor detection, and anchor gating.

Phase-2 implements the full SDE formulation directly, enabling continuous drift adjustment and formal regime classification.

7. EXPERIMENTAL ROADMAP

Planned validation includes bias modulation tests, entropy regularization evaluations, and long-session drift stability analysis.

Response-Bias Correlation (R_b), KL-Divergence (A_{KL}), and Sessional Bias Stability (S_b) will be measured.

8. REPLICATION PLAN

Experiments will run in controlled inference environments using open LLM APIs. Logit distributions will be recorded under adaptive and non-adaptive conditions.

Bootstrap CIs and FDR corrections ensure statistical rigor.

9. CONTRIBUTION SUMMARY

CAAI introduces dynamic memory-conditioned drift, collapse regime detection, and real-time bias regulation—offering a new computational method for stabilizing neural inference.

10. LICENSING & FINGERPRINT

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